



Customer Shopping Behavior Analysis

Business Intelligence & Customer Insights Report — transforming raw transaction data into actionable strategy using Python, PostgreSQL, and Power BI.

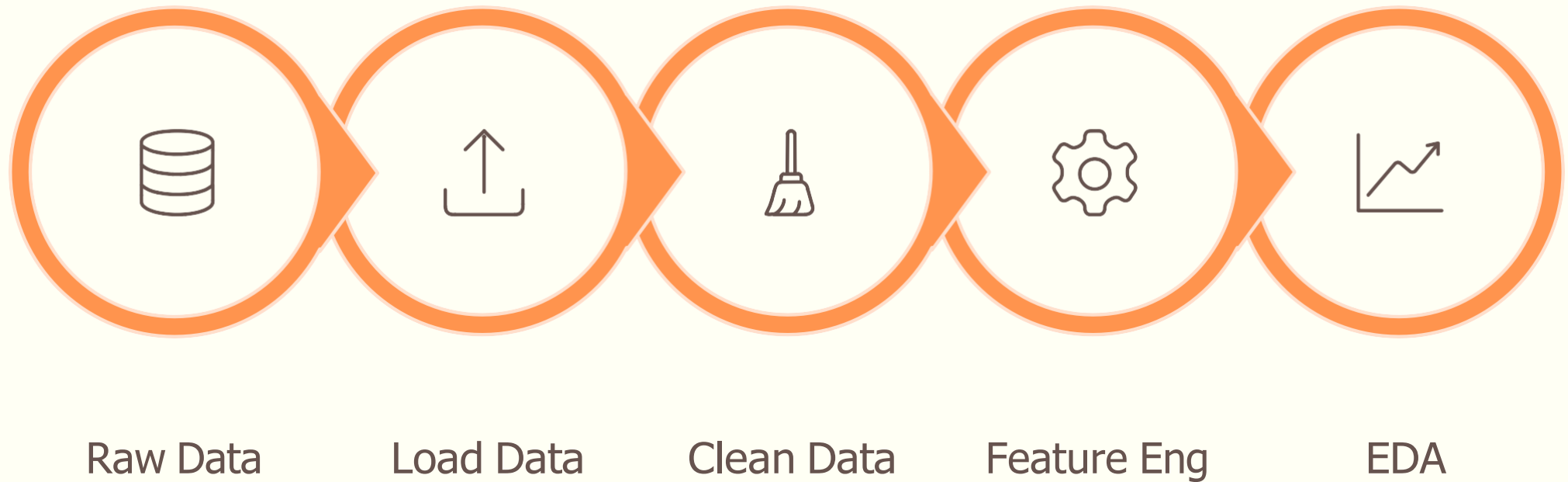
PYTHON · PANDAS · NUMPY

POSTGRESQL · SQL

POWER BI · GAMMA AI

The objective is to convert raw customer transaction data into actionable business insights that support strategic decision-making.

Project Workflow



A structured end-to-end pipeline converting raw transaction data into strategic business recommendations.

Data Loading & Understanding

The dataset was imported into Python using Pandas. Initial inspection used `df.info()`, `df.describe()`, and `df.isnull().sum()` to assess structure, statistics, and missing values.

Dataset Preview

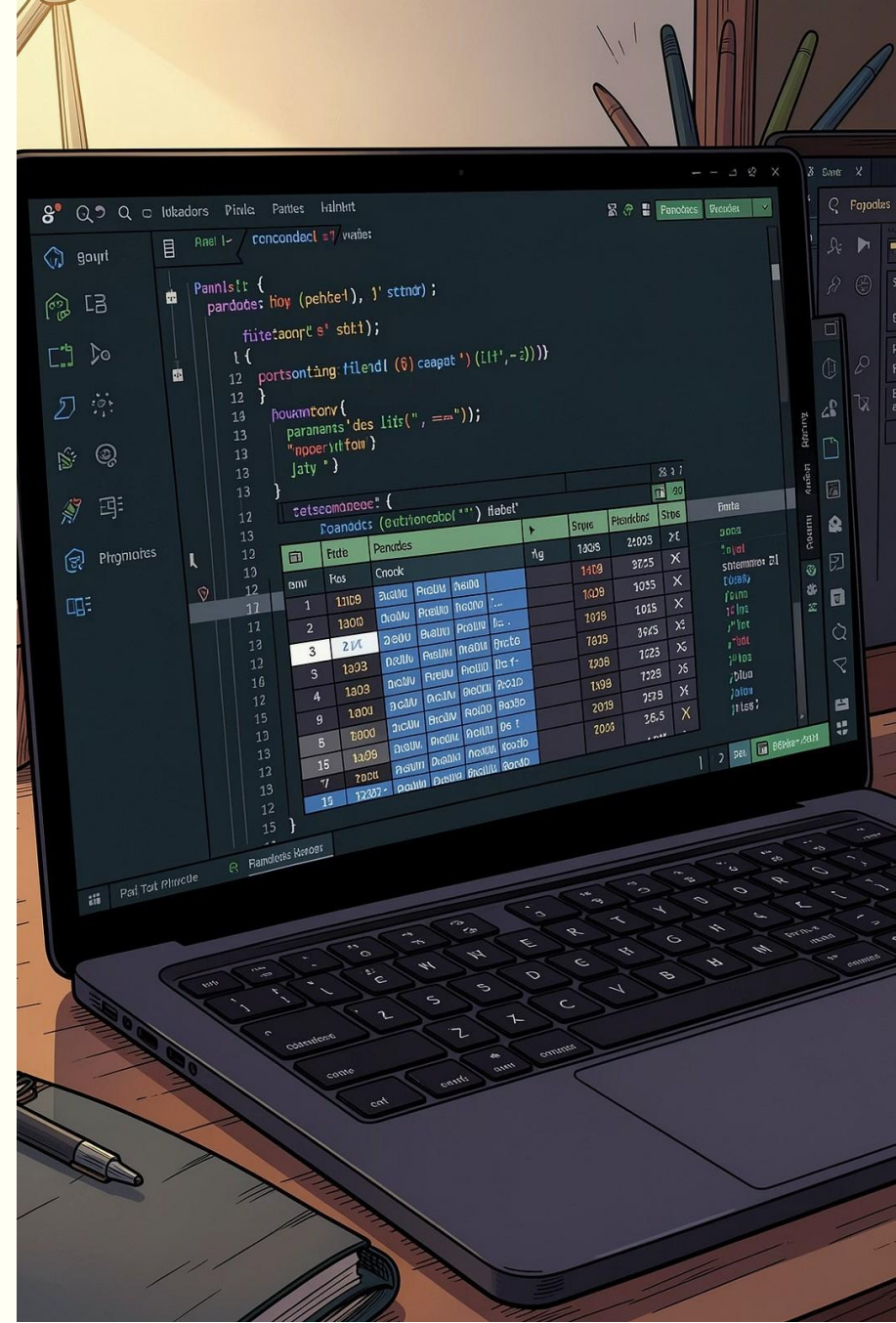
Rows, columns, and data types reviewed

Missing Value Analysis

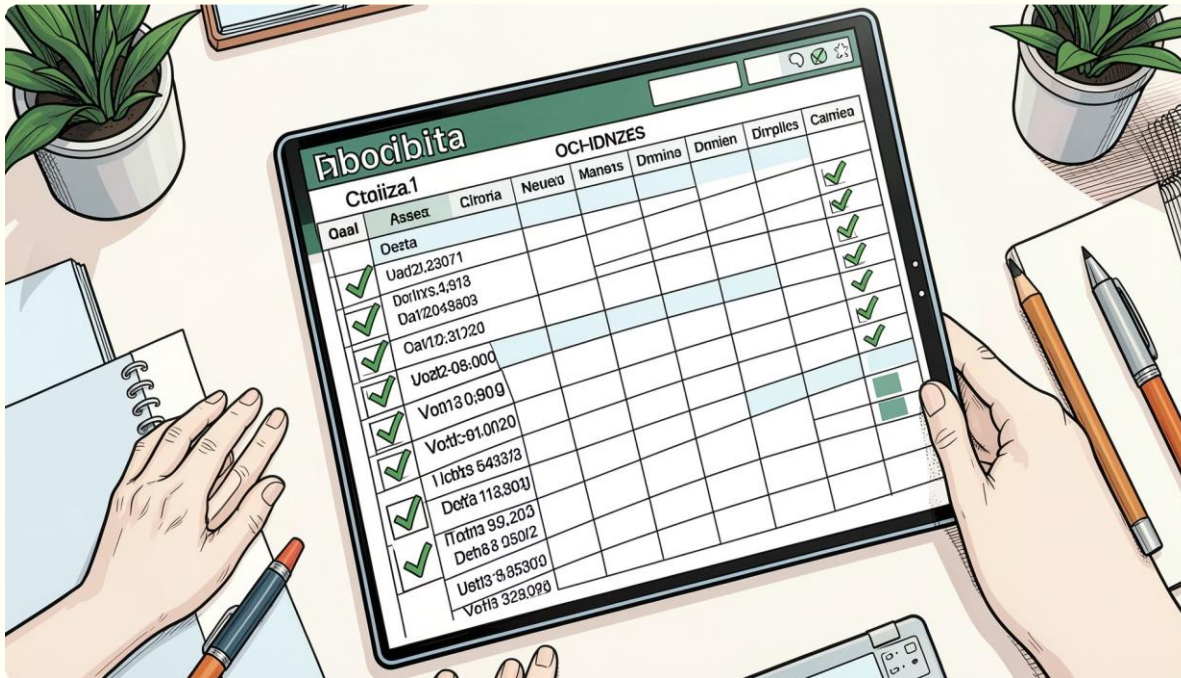
Null counts identified across all fields

Statistical Summary

Distributions and ranges examined



Data Cleaning



Several preprocessing steps were performed to prepare the dataset for analysis.

- **Missing Review Ratings**
Handled null values in rating fields
- **Column Standardization**
Names standardized for consistency
- **Duplicate Variables**
Identified and flagged duplicate fields
- **Irrelevant Fields**
Removed columns not needed for analysis

Feature Engineering

Age Group Segmentation

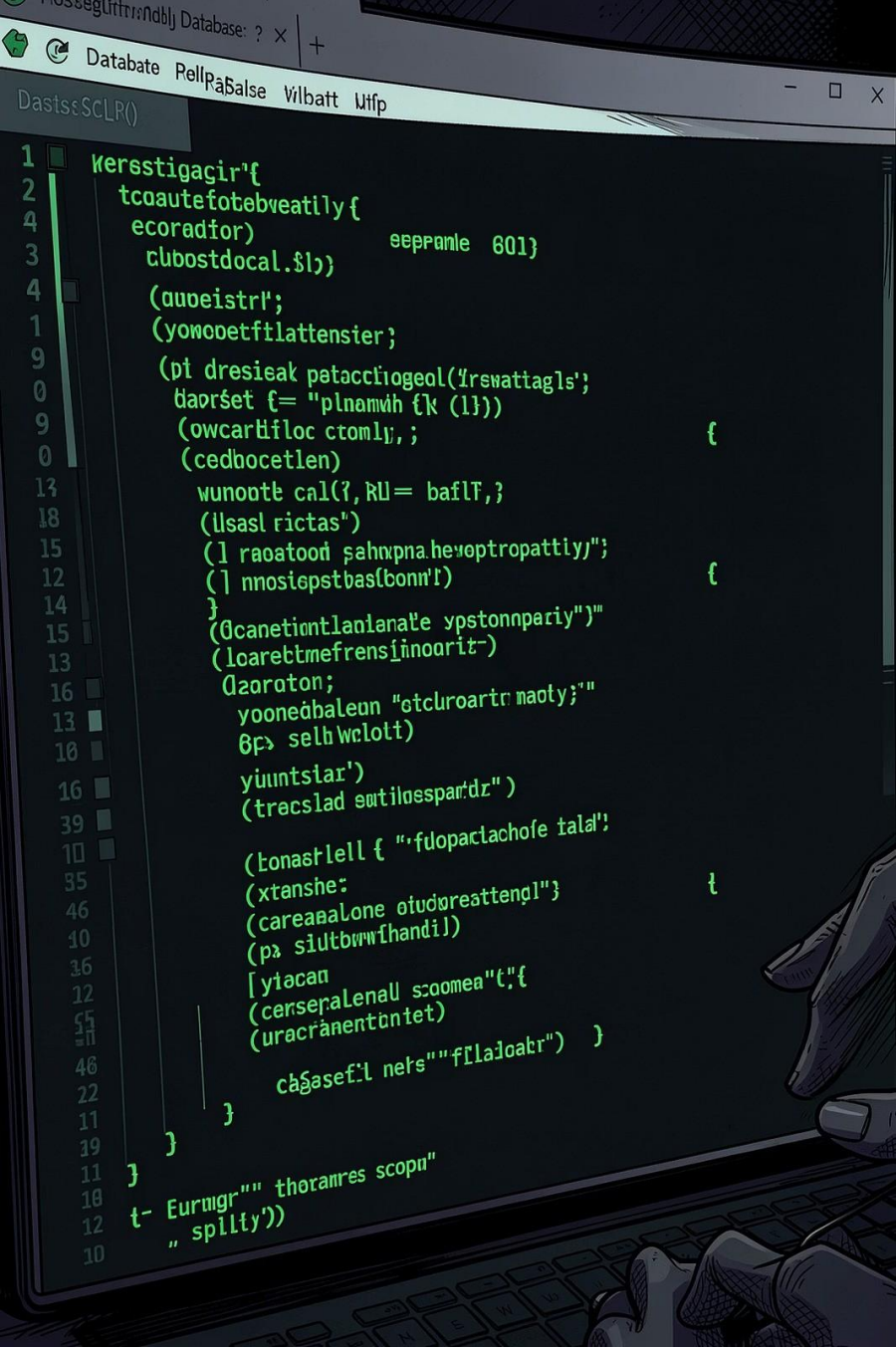
Customers grouped into: Young Adult, Adult, Middle-aged, and Senior segments for demographic analysis.

Purchase Frequency Transformation

Frequency labels converted to numerical days: Weekly → 7, Monthly → 30, Quarterly → 90, Annually → 365.

Data Validation

Identified that **Discount Applied** and **Promo Code Used** contained identical information — redundant field removed.



PostgreSQL & SQL Analysis

After cleaning, the dataset was loaded into PostgreSQL for business analysis.

Revenue by Gender

Male customers generated approximately **2× the revenue** of female customers.

Top-Rated Products

Gloves received the highest average customer rating — a strong-performing category.

Power BI Dashboard & KPIs



Interactive dashboard showing customer segmentation, revenue analysis, sales trends, and subscription behavior.

1.737K

Total Customers

\$60.03

Avg. Purchase

3.72

Avg. Review Rating

Dashboard Findings



Young Adults Lead Revenue

Highest revenue and sales volume — priority demographic for marketing campaigns.



Clothing Dominates Categories

Top sales and revenue contributor — primary focus for promotions and inventory.



Low Subscription Adoption

Only **26.31%** subscribed vs. 73.69% non-subscribed — major opportunity for loyalty programs.

Key Findings & Recommendations



Male Revenue Lead

Significantly higher revenue from male customers



Young Adults

Highest-value customer segment



Clothing Category

Major revenue contributor



Low Subscriptions

Subscription adoption remains low



Gloves Top-Rated

Highest customer satisfaction rating

Boost Subscriptions

Loyalty rewards, exclusive discounts, membership benefits

Target Young Adults

Personalized promotions, social campaigns, recommendations

Promote Top Products

Featured listings, bundled offers, upselling

Improve Retention

Repeat customer rewards, referral programs, engagement